

Exam**August 16th, 2024****Model Predictive Control (151-0660-00)****Prof. M. Zeilinger**

Exam Question Booklet

Please provide your student number and write your first and last name below.

Student number:

.....

First and last name:

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Exam Duration:	120 Min
Number of Questions:	10
Number of Points:	100
Permitted Aids:	Two A4 sheets of paper (4 pages, handwritten or computer typed)
Important:	No negative points will be given for incorrect answers.

Answers should be given in the provided Answer booklet, or in the Question booklet, if required by the question.

Use only the provided Answer booklet for your calculations; additional paper is available from the supervisors.

For each multiple choice question you are given four statements and you have to decide which of them are correct. For correct answers to all four statements the full number of points is allocated, for three correct answers half the number of points, and zero points otherwise.

Good luck!

S Supplementary Material

Definitions

Definitions used throughout the exam, which are for your reference and correspond to the definitions used in the lecture:

$$\text{State sequence} \quad X := \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{bmatrix}, x_i \in \mathbb{R}^n$$

$$\text{Input sequence} \quad U := \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}, u_i \in \mathbb{R}^m$$

$$\text{Nominal state sequence} \quad Z := \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{bmatrix}, z_i \in \mathbb{R}^n$$

$$\text{Nominal input sequence} \quad V := \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}, v_i \in \mathbb{R}^m$$

$$\text{One vector} \quad \mathbf{1}_n := [1, 1, \dots, 1]^T \in \mathbb{R}^n$$

$$\text{Identity matrix} \quad \mathbb{I}_{n \times n} := \text{diag}(\mathbf{1}_n) \in \mathbb{R}^{n \times n}$$

$$\text{Open interval} \quad (-a, a) := \{x \in \mathbb{R} \mid -a < x < a\}$$

$$\text{Closed interval} \quad [-a, a] := \{x \in \mathbb{R} \mid -a \leq x \leq a\}$$

$$\text{1-norm} \quad \|x\|_1 = \|[x_1, \dots, x_n]^T\|_1 := \sum_i |x_i|$$

$$\infty\text{-norm} \quad \|x\|_\infty = \|[x_1, \dots, x_n]^T\|_\infty := \max_i (|x_i|)$$

$$\text{2-norm} \quad \|x\|_2 = \|[x_1, \dots, x_n]^T\|_2 := \sqrt{\sum_i x_i^2} = \sqrt{x^T x}$$

$$\text{Weighted 2-norm} \quad \|x\|_Q = \|[x_1, \dots, x_n]^T\|_Q := \sqrt{x^T Q x}$$

$$\begin{aligned} \text{Minkowski sum} \quad \mathcal{A}_1 \oplus \mathcal{A}_2 &= \{x_1 + x_2 \mid x_1 \in \mathcal{A}_1, x_2 \in \mathcal{A}_2\} \\ \text{(scalar case)} \quad [a, b] \oplus [c, d] &= [a + c, b + d] \end{aligned}$$

$$\begin{aligned} \text{Pontryagin difference} \quad \mathcal{A}_1 \ominus \mathcal{A}_2 &= \{x_1 \mid x_1 + x_2 \in \mathcal{A}_1 \forall x_2 \in \mathcal{A}_2\} \\ \text{(scalar case)} \quad [a, b] \ominus [c, d] &= [a - c, b - d] \end{aligned}$$

$$\text{Union} \quad \mathcal{A}_1 \cup \mathcal{A}_2 = \{x \mid x \in \mathcal{A}_1 \text{ or } x \in \mathcal{A}_2\}$$

$$\text{Intersection} \quad \mathcal{A}_1 \cap \mathcal{A}_2 = \{x \mid x \in \mathcal{A}_1, x \in \mathcal{A}_2\}$$

$$\text{Set difference} \quad \mathcal{A}_1 \setminus \mathcal{A}_2 = \{x \mid x \in \mathcal{A}_1, x \notin \mathcal{A}_2\}$$

- If not specified otherwise, an MPC control law refers to $\kappa(x(k)) = u_0^*(x(k))$, i.e. the first element of the optimal input sequence computed by solving the associated optimization problem initialized at $x_0 = x(k)$.

S.1 Standard Nominal MPC Problem

Consider linear systems of the form

$$x(k+1) = Ax(k) + Bu(k). \quad (1)$$

The standard linear MPC control law $\kappa(x(k)) = u_0^*(x(k))$ is defined by

$$\begin{aligned} P_{S.1} : \quad J^*(x(k)) &= \min_U l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i) \\ \text{s.t.} \quad x_{i+1} &= Ax_i + Bu_i, \quad i = 0, \dots, N-1, & (2a) \\ x_i &\in \mathcal{X}, \quad i = 0, \dots, N-1, & (2b) \\ u_i &\in \mathcal{U}, \quad i = 0, \dots, N-1, & (2c) \\ x_N &\in \mathcal{X}_f, & (2d) \\ x_0 &= x(k). & (2e) \end{aligned}$$

Assumption S.1-1 The stage cost $l(x, u)$ is positive definite and only zero at the origin.

Assumption S.1-2

- The terminal set $\mathcal{X}_f$ is a positively invariant set for system (1) under the controller $\kappa_f(x)$:

$$Ax + B\kappa_f(x) \in \mathcal{X}_f, \quad \text{for all } x \in \mathcal{X}_f.$$

- All state and input constraints are satisfied in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X}, \quad \kappa_f(x) \in \mathcal{U}, \quad \text{for all } x \in \mathcal{X}_f.$$

Assumption S.1-3 The terminal cost is a continuous Lyapunov function in the terminal set $\mathcal{X}_f$ and satisfies:

$$l_f(Ax + B\kappa_f(x)) - l_f(x) \leq -l(x, \kappa_f(x)), \quad \text{for all } x \in \mathcal{X}_f.$$

S.2 Standard Tube-based MPC Problem

Consider uncertain linear systems of the form

$$x(k+1) = Ax(k) + Bu(k) + w(k) \quad (3)$$

with additive disturbances $w(k) \in \mathcal{W}$, where $\mathcal{W}$ is a compact set containing the origin.

The standard tube-based MPC control law $\kappa_{\text{tube}}(x(k)) = K_{\mathcal{E}}(x(k) - z_0^*(x(k))) + v_0^*(x(k))$ is defined by

$$P_{S.2} : \quad \tilde{J}^*(x(k)) = \min_{V, z_0} l_f(z_N) + \sum_{i=0}^{N-1} l(z_i, v_i)$$

$$\text{s.t.} \quad z_{i+1} = Az_i + Bv_i, \quad i = 0, \dots, N-1, \quad (4a)$$

$$z_i \in \mathcal{X} \ominus \mathcal{E}, \quad i = 0, \dots, N-1, \quad (4b)$$

$$v_i \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \quad i = 0, \dots, N-1, \quad (4c)$$

$$z_N \in \tilde{\mathcal{X}}_f, \quad (4d)$$

$$x(k) \in z_0 \oplus \mathcal{E}. \quad (4e)$$

Assumption S.2-1 The stage cost $l(z, v)$ is positive definite and only zero at the origin.

Assumption S.2-2 The set $\mathcal{E}$ is a robust invariant set for system (3) under $u(k) = K_{\mathcal{E}}x(k)$.

Assumption S.2-3

- The terminal set $\tilde{\mathcal{X}}_f$ is a positively invariant set for the nominal system $z(k+1) = Az(k) + Bv(k)$ under the controller $\tilde{\kappa}_f(z(k))$:

$$Az + B\tilde{\kappa}_f(z) \in \tilde{\mathcal{X}}_f, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

All tightened state and input constraints are satisfied in $\tilde{\mathcal{X}}_f$:

$$\tilde{\mathcal{X}}_f \subseteq \mathcal{X} \ominus \mathcal{E}, \quad \tilde{\kappa}_f(z) \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}, \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

- The terminal cost is a continuous Lyapunov function in the terminal set $\tilde{\mathcal{X}}_f$ and satisfies:

$$l_f(Az + B\tilde{\kappa}_f(z)) - l_f(z) \leq -l(z, \tilde{\kappa}_f(z)), \quad \text{for all } z \in \tilde{\mathcal{X}}_f.$$

Question 1

[3 pt. | 4 pt.]

Consider the following nonlinear discrete-time system

$$x(k+1) = f(x(k)), \tag{5}$$

with the state $x(k) \in \mathbb{R}$.

- i) Assume $f(x) = 2x - x^3$ and compute all the equilibrium points of system (5). Using Lyapunov's indirect method, what can be concluded about their local stability?
- ii) Assume $f(x) = x - x^5$ and consider the Lyapunov candidate function $V(x) = x^2$. Prove asymptotic stability of the origin ($x = 0$) for system (5) in the set $\Omega = [-1, 1]$.

Question 2

[2pt. | 3pt. | 4pt. | 2pt.]

Consider the optimization problem

$$P_O : \quad p^* = \min_{x \in \mathbb{R}^2} x_1^2 + x_2^2, \\ \text{s.t.} \quad x_2 - x_1 \leq \beta,$$

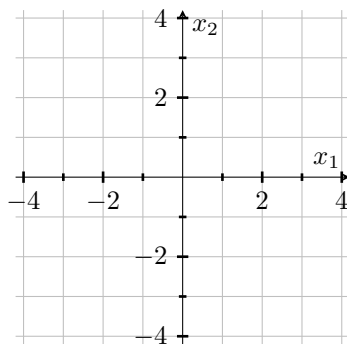
where $\beta \in \mathbb{R}$.

- i) Show for the optimization problem P_O that the optimal cost of the dual problem is equal to the optimal cost of the primal problem, i.e., $d^* = p^*$. In your solution, state the corresponding sufficient conditions and show that they are satisfied.

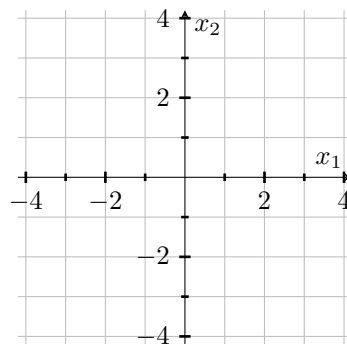
Hint: You do not have to compute p^ and d^* .*

- ii) State the Lagrangian $L(x, \lambda)$ and the Karush-Kuhn-Tucker (KKT) optimality conditions for P_O .
- iii) State and solve the dual problem of P_O . Provide expressions for the dual optimizer λ^* and dual cost d^* as a function of β .
- iv) Let $\beta = -2$. Draw the feasible set for P_O and the contour line $\{x \mid x_1^2 + x_2^2 = p^*\}$ into the left diagram below. State the primal optimizer x^* and mark it in the left diagram. You may use the right diagram for sketching.

Answer



Sketch



Question 3

[3 pt. | 3 pt.]

Consider the following constrained finite-time optimal control (CFTOC) problem

$$\begin{aligned} J_p^*(x(k)) = \min_U \quad & J_p(X, U) \\ \text{s.t.} \quad & x_{i+1} = Ax_i + Bu_i, \quad \forall i = 0, \dots, N-1, \\ & Gu_i \leq h, \quad \forall i = 0, \dots, N-1, \\ & x_0 = x(k), \end{aligned} \tag{6}$$

with state $x(k), x_0, \dots, x_N \in \mathbb{R}^n$, input $u_0, \dots, u_{N-1} \in \mathbb{R}^m$, and possible cost functions

$$J_1(X, U) = c_x^\top x_N + \sum_{i=0}^{N-1} c_x^\top x_i + c_u^\top u_i, \quad \text{with } c_x \in \mathbb{R}^n \text{ and } c_u \in \mathbb{R}^m, \tag{7}$$

$$J_2(X, U) = x_N^\top x_N + \sum_{i=0}^{N-1} x_i^\top x_i + u_i^\top u_i, \tag{8}$$

$$J_\infty(X, U) = \|x_N\|_\infty + \sum_{i=0}^{N-1} \|x_i\|_\infty + \|u_i\|_\infty. \tag{9}$$

i) Mark all statements that are guaranteed to be true:

	True
The feasible set of problem (6) when $p = 1$ differs from the feasible set of problem (6) when $p = \infty$.	☐
The optimal solution X^*, U^* to problem (6) with $p = 2$ is unique.	☐
The optimal cost $J_1^*(x(k))$ to problem (6) with $p = 1$ is bounded.	☐
The problem (6) with $p = 2$ solved in receding horizon is feasible $\forall x(k) \in \mathbb{R}^n$ with $0 \leq k \leq \bar{k}$, where $\bar{k}$ is a finite constant.	☐

ii) Mark all statements that are guaranteed to be true:

	True
The dual problem to problem (6) with $p = 1$ is an LP.	☐
The optimal solution $u_0^* = \kappa(x(k))$ to problem (6) with $p = \infty$ is a piecewise affine function of the state $x(k)$ defined over polyhedral sets that partition the feasible set.	☐
The dual function to problem (6) is concave for any $p \in \{1, 2, \infty\}$.	☐
The optimal cost $J_2^*(x(k))$ to problem (6) with $p = 2$ is a convex and piecewise affine function of the state $x(k)$ defined over polyhedral sets that partition the feasible set.	☐

Question 4

[3pt. | 3pt. | 3pt.]

Consider the linear time-invariant system

$$x(k+1) = Ax(k) + Bu(k) \quad (10)$$

and the standard MPC control law $\kappa(x(k)) = u_0^*(x(k))$ as defined in the Supplementary Material by optimization problem $P_{S,1}$, satisfying Assumptions S.1-1, S.1-2 and S.1-3.

- i) Let $x_{\text{MPC}}(0), x_{\text{MPC}}(1), \dots$ be the sequence of states of the closed-loop system under the MPC control law and $\mathcal{X}_N$ define the feasible set. Assume that $P_{S,1}$ is feasible for $x_{\text{MPC}}(0)$.

Mark all statements that are guaranteed to be true.

	True
$\lim_{k \rightarrow \infty} x_{\text{MPC}}(k) = 0$	☐
$J^*(x_{\text{MPC}}(k)) \leq J^*(x_{\text{MPC}}(0))$	☐
$x_{\text{MPC}}(k) \in \mathcal{X}_N$ for all $k \geq 0$	☐
If $x_{\text{MPC}}(k) \in \mathcal{X}_f$ then $x_{\text{MPC}}(k+1) \in \mathcal{X}_f$ for all $k \geq 0$	☐

- ii) Consider a positive invariant set $\mathcal{O} \subseteq \mathcal{X}$ for system (10) under a controller $\kappa_f(x) \in \mathcal{U} \forall x \in \mathcal{O}$. Consider also the maximum control invariant set $\mathcal{C}_\infty$ of system (10) subject to constraints $x(k) \in \mathcal{X}$ and $u(k) \in \mathcal{U}$, i.e., (2b) and (2c) in $P_{S,1}$. We consider $P_{S,1}$ with terminal set $\mathcal{X}_f = \mathcal{O}$, and denote the feasible set $\mathcal{X}_N^o$ and the predicted open-loop cost as $J_o^*(x(0))$. Similarly, for $\mathcal{X}_f = \mathcal{C}_\infty$, we define $\mathcal{X}_N^c$ and $J_c^*(x(0))$.

Mark all statements that are guaranteed to be true.

	True
$\mathcal{X}_N^o \subseteq \mathcal{X}_N^c$	☐
$\mathcal{O} \subseteq \mathcal{X}_N^c$	☐
$\mathcal{X}_N^c \subseteq \mathcal{C}_\infty$	☐
$J_o^*(x(0)) \leq J_c^*(x(0))$	☐

- iii) Consider $\kappa_f(x) = 0$ and let the system $x(k+1) = Ax(k)$ be asymptotically stable. Additionally, consider $Q \succ 0$, no state constraints, i.e., $\mathcal{X} = \mathbb{R}^n$, and that $\mathcal{U}$ contains the origin. Finally, assume a terminal cost function of the form $l_f(x) = x^\top Px$, and a terminal set $\mathcal{X}_f := \{x \mid x^\top Px \leq \alpha\}$ for some $\alpha > 0$.

Mark the conditions under which Assumption S.1-2. is satisfied.

	True
$A^\top PA - P \prec -Q$	☐
$A^\top PA - P \prec 0$	☐
$A^\top PA - P \prec -a\mathbb{I}$ for any $a \in \mathbb{R}$	☐
$P \succ 0$	☐

Question 5

[3pt. | 2pt. | 4pt. | 4pt.]

Consider the linear time-invariant system

$$x(k+1) = Ax(k) + Bu(k), \quad (11)$$

with $x(k) \in \mathbb{R}^n$ and $u(k) \in \mathbb{R}^m$. Further, consider the MPC control law defined by the following optimization problem

$$\begin{aligned} U^*(x(k)) = \arg \min_U \quad & x_N^\top P x_N + \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i \\ \text{s.t.} \quad & x_{i+1} = Ax_i + Bu_i, \quad i = 0, \dots, N-1, \\ & x_0 = x(k), \end{aligned} \quad (12)$$

with positive definite matrices P , Q , and R .

i) Mark the statements that are guaranteed to be true:

	True
If the origin is locally asymptotically stable for the system (11) under a linear controller $u(k) = Fx(k)$, then it is also globally asymptotically stable under this controller.	☐
Suppose the origin is globally asymptotically stable for the system (11) under the control law obtained by solving problem (12) in receding horizon for $N = \bar{N}$. Then, the origin is globally asymptotically stable for the same system under the control law obtained by solving problem (12) in receding horizon for $N = \bar{N} + 1$.	☐
If P in problem (12) is chosen as the solution of $A^\top P A + Q = P$, the closed-loop system (11) under the control law $u(k) = u_0^*(x(k))$ obtained by solving (12) is guaranteed to be globally asymptotically stable.	☐
The control law obtained by solving problem (12) in receding horizon at every $x(k)$ is of the form $u(k) = Fx(k)$.	☐

Consider the following scalar discrete-time system

$$x(k+1) = 2x(k) + u(k). \quad (13)$$

The MPC control law is defined by

$$\begin{aligned} U^*(x(k)) = \arg \min_U \quad & p\|x_2\|^2 + \sum_{i=0}^1 \|x_i\|^2 + r\|u_i\|^2 \\ \text{s.t.} \quad & x_{i+1} = 2x_i + u_i, \quad i = 0, 1, \\ & x_0 = x(k), \end{aligned} \quad (14)$$

with scalar cost weights $0 < r \in \mathbb{R}$ and $0 < p \in \mathbb{R}$.

- ii) For $p = 1$ and $r = 5$, the optimal input sequence is $U^*(x(k)) = \left[-\frac{13x(k)}{14}, -\frac{5x(k)}{14}\right]^\top$. Is the resulting closed-loop system under the MPC control law $\kappa_{\text{MPC}}(x(k)) = u_0^*(x(k))$ asymptotically stable? Justify your answer.
- iii) For $p = 0$ and any $r > 0$, consider a modification of the MPC problem (14) by adding the zero terminal constraint $x_2 = 0$. Is the closed-loop system starting at $x(0) = 2$ asymptotically stable under the modified MPC control law? Justify your answer.
- iv) Consider a modification of the MPC problem (14) by adding an additional constraint $\|u_i\|_\infty \leq 1$ for $i = 0, 1$. Given a linear terminal controller $\kappa_f(x) = -1.5x$, design the largest terminal set $\mathcal{X}_f$ such that the MPC problem (14) with the terminal constraint $x_2 \in \mathcal{X}_f$ satisfies the Assumption S1.2.

Question 6

[4 pt. | 4 pt. | 4 pt.]

Consider the linear time-invariant system

$$x(k+1) = Ax(k) + Bu(k), \quad (15)$$

with $x(k) \in \mathbb{R}^n$ and $u(k) \in \mathbb{R}^m$, and constraints $x(k) \in \mathcal{X} \subseteq \mathbb{R}^n$ and $u(k) \in \mathcal{U} \subseteq \mathbb{R}^m$ with $\mathcal{X}, \mathcal{U}$ convex and both containing the origin in the interior. Assume we have access to a state and input trajectory

$$\mathcal{S} := \{\tilde{x}_0, \dots, \tilde{x}_T\}, \mathcal{V} := \{\tilde{u}_0, \dots, \tilde{u}_{T-1}\}, \quad (16)$$

ending at the origin, i.e., $\tilde{x}_T = 0$, satisfying constraints, i.e., $\tilde{x}_j \in \mathcal{X}$ and $\tilde{u}_j \in \mathcal{U}$, and the dynamics (15), i.e., $\tilde{x}_{j+1} = A\tilde{x}_j + B\tilde{u}_j$, for all $j = 0, \dots, T-1$.

Consider the following MPC controller

$$J_N^*(x(k)) = \min_{X, U} l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i) \quad (17a)$$

$$\text{s.t. } x_{i+1} = Ax_i + Bu_i, \quad i = 0, \dots, N-1, \quad (17b)$$

$$x_i \in \mathcal{X}, \quad i = 0, \dots, N-1, \quad (17c)$$

$$u_i \in \mathcal{U}, \quad i = 0, \dots, N-1, \quad (17d)$$

$$x_N \in \mathcal{S}, \quad (17e)$$

$$x_0 = x(k), \quad (17f)$$

with positive definite stage cost $l(x, u)$ and terminal cost $l_f : \mathcal{S} \mapsto \mathbb{R}$ defined as

$$l_f(\tilde{x}_j) = \sum_{i=j}^{T-1} l(\tilde{x}_i, \tilde{u}_i), \quad \forall j = 0, \dots, T-1, \quad (18)$$

and $l_f(\tilde{x}_T) = l_f(0) = 0$.

- i) Prove that the MPC problem (17) is recursively feasible.
- ii) Prove that the origin is asymptotically stable for the closed-loop system (15) under the MPC controller resulting from (17) in the feasible set $\mathcal{X}_N = \{x(k) \mid (17b) - (17f)\}$.
- iii) Show that the convex hull $\text{co}(\mathcal{S})$ is a control invariant set for system (15), i.e., for any $x \in \text{co}(\mathcal{S})$ there exists a $u \in \mathcal{U}$ such that $Ax + Bu \in \text{co}(\mathcal{S})$.

Question 7

[2pt. | 2pt. | 5pt. | 4pt.]

Consider the following scalar linear time-invariant system

$$x(k+1) = \frac{3}{2}x(k) + u(k) + d_\infty, \quad (19a)$$

$$y(k) = x(k) + d_\infty, \quad (19b)$$

$$x(k) \in \mathcal{X} = [-2, 2], \quad (19c)$$

$$u(k) \in \mathcal{U} = [-2, 2], \quad (19d)$$

and the following MPC problem for tracking in delta formulation

$$\min_{\Delta x_i, \Delta u_i} \sum_{i=0}^{N-1} \Delta x_i^T Q \Delta x_i + \Delta u_i^T R \Delta u_i + V_f(\Delta x_N) \quad (20a)$$

$$\text{s.t. } \Delta x_0 = \Delta x(k), \quad (20b)$$

$$\Delta x_{i+1} = A \Delta x_i + B \Delta u_i, \quad (20c)$$

$$G_{\Delta x} \Delta x_i \leq g_{\Delta x}, \quad (20d)$$

$$G_{\Delta u} \Delta u_i \leq g_{\Delta u}, \quad (20e)$$

$$\Delta x_N \in \mathcal{X}_f, \quad (20f)$$

where $\Delta x(k) = x(k) - x_s$ and $\Delta u(k) = u(k) - u_s$ are given in terms of the steady state x_s and corresponding control input u_s , and the applied control input to system (19a) is defined as

$$u(k) = u_s + \Delta u_0^*. \quad (21)$$

- i) Let (x_s, u_s) be the steady-state pair such that the output at the steady state is equal to the reference, i.e., $y_s = r$. State x_s and u_s as a function of d_∞ and r .

From here on, consider the reference $r = 1$ with steady state $x_s = 1, u_s = -\frac{1}{2}$ for $d_\infty = 0$.

- ii) Provide the numerical values of $G_{\Delta x}$, $g_{\Delta x}$, $G_{\Delta u}$ and $g_{\Delta u}$ such that

- $G_{\Delta x} \Delta x(k) \leq g_{\Delta x} \Leftrightarrow x(k) \in \mathcal{X}$, and
- $G_{\Delta u} \Delta u(k) \leq g_{\Delta u} \Leftrightarrow u(k) \in \mathcal{U}$.

- iii) Provide a terminal set $\mathcal{X}_f = [\Delta x_{\min}, \Delta x_{\max}]$ for which

- Assumption S.1-2 in the Supplementary Material is satisfied under the terminal control law $\Delta u(k) = \kappa_f(\Delta x(k)) = (-1) \cdot \Delta x(k)$,
- the MPC controller is feasible at $x(k) = -2$ for $N = 1$.

Hint: It is not required to derive the MPC control law.

- iv) Let $Q = 1$, $R = 1$. Consider the terminal set $\mathcal{X}_f = [-1, 1]$, for which Assumption S.1-2 in the Supplementary Material holds (with x replaced by Δx) under the terminal controller $\kappa_f(\Delta x) = (-1) \cdot \Delta x$. Suppose that the MPC controller is feasible at $\Delta x(0)$. Provide a terminal cost $V_f(\Delta x_N)$ for which the closed-loop system under the MPC control law is guaranteed to converge to the target reference r , i.e., $y(k) \rightarrow r$ for $k \rightarrow \infty$. Justify your answer.

Question 8

[2pt. | 4pt. | 1pt. | 3pt. | 4pt.]

Consider the uncertain system

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} \sin(x_2(k) + \varphi_1) + x_2(k) \\ 2 \cos^2(x_1(k) + \varphi_2) \end{bmatrix} + u(k) + w_\varphi(k) \quad (22)$$

where $u(k) \in \mathbb{R}^2$ is the input, $\varphi_1, \varphi_2 \in [0, 2\pi]$ are unknown parameters, and $w_\varphi(k) \in \mathcal{W}_\varphi$ a disturbance with $\mathcal{W}_\varphi = \{w \in \mathbb{R}^2 \mid \|w\|_\infty \leq 0.1\}$. The system is subject to state and input constraints

$$\mathcal{X} = \{x \in \mathbb{R}^2 \mid \|x\|_\infty \leq 2\}, \quad (23)$$

$$\mathcal{U} = \{u \in \mathbb{R}^2 \mid \|u\|_\infty \leq 2\}. \quad (24)$$

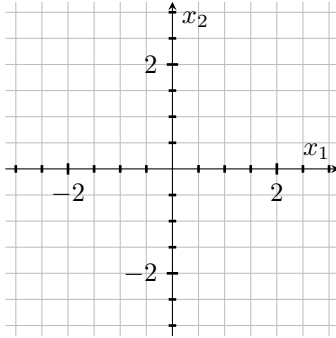
i) Provide the smallest disturbance set $\mathcal{W} \subset \mathbb{R}^2$ such that the system

$$x(k+1) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(k) + u(k) + w(k) \quad (25)$$

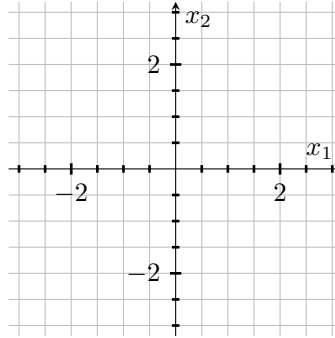
with $w(k) \in \mathcal{W}$ captures all possible trajectories of system (22) starting from the same initial condition.

From here on, assume that $\mathcal{W} = \{w \in \mathbb{R}^2 \mid \|w\|_\infty \leq 0.5\}$.

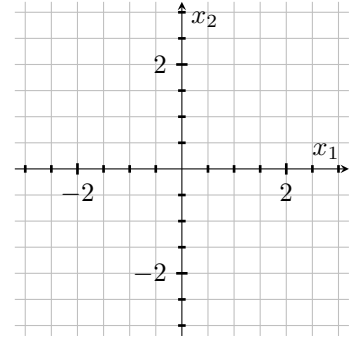
ii) Compute the disturbance-reachable sets $\mathcal{F}_k$ of the system (25) under the control inputs $u(k) = \begin{bmatrix} 0 & 0 \end{bmatrix}^\top$, starting from $\mathcal{F}_0 = \{0\}$, for $k \in \{1, 2, 3\}$. Draw them into the diagrams below:



a) $k = 1$



b) $k = 2$



c) $k = 3$

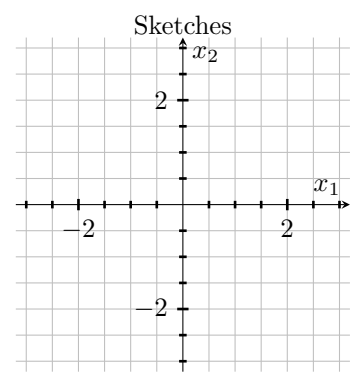
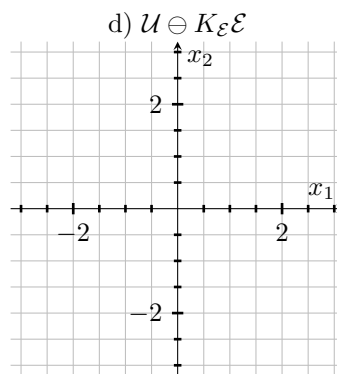
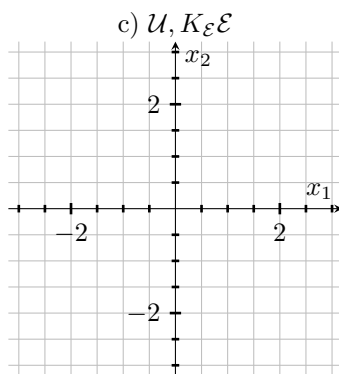
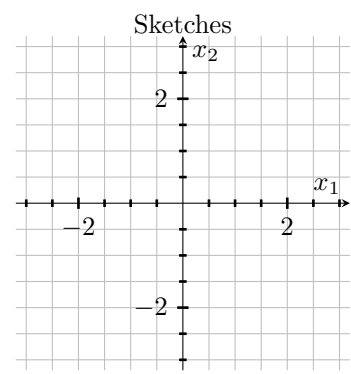
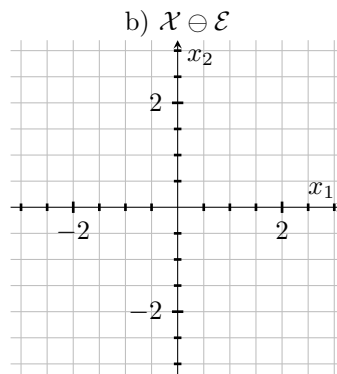
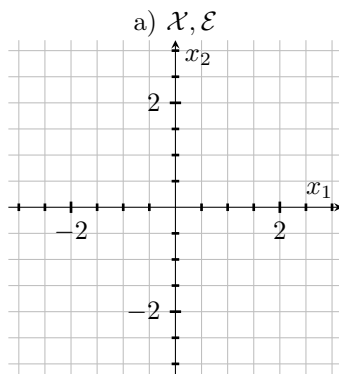
iii) State the minimum robust positive invariant set of system (25) under the control law $u(k) = \begin{bmatrix} 0 & 0 \end{bmatrix}^\top$.
Hint: There is no derivation needed to answer this question.

Consider the tube-MPC problem $P_{S,2}$ in the Supplementary Material, with

- the set $\mathcal{E} = \{e \in \mathbb{R}^2 \mid \|e\|_\infty \leq 1\}$, and
- the linear control law $u(k) = K_{\mathcal{E}}x(k)$, with $K_{\mathcal{E}} = \begin{bmatrix} -0.5 & -1 \\ 0 & -0.5 \end{bmatrix}$.

iv) Show that $\mathcal{E}$ is a robust positive invariant set of system (25) under the control law $u(k) = K_{\mathcal{E}}x(k)$.

- v) Draw the following sets in the diagrams below (six sets in total). For a) and c), label the sets with their respective name. You may use the right-most diagrams for sketches.



Question 9

[2pt. | 3pt. | 2pt.]

Consider the tube-based MPC scheme as presented in the Supplementary Material S.2 and assume that Assumptions S.2-1, S.2-2, and S.2-3 hold. Consider the following modifications and state if the resulting MPC problem is guaranteed to be recursively feasible. Justify your answers.

- i) Remove the constraint $z_0 \in \mathcal{X} \ominus \mathcal{E}$ in (4b).
- ii) Replace $x(k) \in z_0 \oplus \mathcal{E}$ in (4e) by $x(k) = z_0$.
- iii) Replace $\mathcal{E}$ in the optimization problem $P_{S.2}$ by $2\mathcal{E}$. Assume that the modified problem is feasible at $k = 0$.

Question 10

[4pt. | 4pt.]

Consider the uncertain linear time-invariant system

$$x(k+1) = Ax(k) + Bu(k) + w(k) \quad (26)$$

with $w(k) \in \mathcal{W}$ and assume that $\mathcal{W}$ is compact and contains the origin in its interior.

i) Mark all the statements that are guaranteed to be true.

	True
If $x(k+1) = Ax(k)$ is asymptotically stable, a closed and bounded robust invariant set $\mathcal{E}$ for (26) exists under the controller $u(k) = 0$.	☐
If the sets $\mathcal{X}$ and $\mathcal{E}$ are polytopes, then the set $\mathcal{X} \ominus \mathcal{E}$ is a polytope.	☐
If the sets $\mathcal{E}_1$ and $\mathcal{E}_2$ both satisfy Assumption S.2-2 under $u(k) = K_{\mathcal{E}}x(k)$, then the set defined as the intersection of $\mathcal{E}_1$ and $\mathcal{E}_2$, i.e. $\mathcal{E}_1 \cap \mathcal{E}_2$, also satisfies Assumption S.2-2 under $u(k) = K_{\mathcal{E}}x(k)$.	☐
Let $\mathcal{E}_{\min}$ be the minimum robust positive invariant set for (26) under the controller $u(k) = K_{\mathcal{E}}x(k)$, then it holds that $\bigoplus_{j=0}^{\infty} A^j \mathcal{W} = \mathcal{E}_{\min}$.	☐

Let system (26) be controlled by the standard tube-based MPC control law $\kappa_{\text{tube}}(x(k))$ as presented in the Supplementary Material S.2, with $P_{S.2}$ satisfying the Assumptions S.2-1, S.2-2, and S.2-3. Let $P_{S.2}$ be feasible for $x(0)$.

ii) Mark all the statements that are guaranteed to be true.

	True
For the closed-loop system (26) under the tube-based MPC controller $u(k) = \kappa_{\text{tube}}(x(k))$ it holds that $\lim_{k \rightarrow \infty} x(k) = 0$.	☐
The closed-loop input is contained within the tightened input constraints, i.e., $\kappa_{\text{tube}}(x(k)) \in \mathcal{U} \ominus K_{\mathcal{E}}\mathcal{E}$ for all $k \geq 0$.	☐
For the closed-loop system (26) under the tube-based MPC controller $u(k) = \kappa_{\text{tube}}(x(k))$ it holds that $x(N) \in \tilde{\mathcal{X}}_f \oplus \mathcal{E}$.	☐
The infinite-horizon cost corresponding to the optimal nominal state $z_0^*(x(k))$ and control input $v_0^*(x(k))$ is bounded, i.e., $\sum_{k=0}^{\infty} l(z_0^*(x(k)), v_0^*(x(k))) < \infty$.	☐